CREST AWARDS | Student profile

First name *Please provide only the first name of the student/team member. Do not include middle or last names.*	Devadit
CREST Award Level	Gold
Project title	Behavioural Fingerprinting from Keystroke Dynamics
Mentor name *(if you had one)*	-

This Profile form is suitable for Bronze, Silver and Gold CREST Awards. Always use in conjunction with corresponding Student Guide Documents.

Please fill in this profile form and submit it along with your project work or report. This a requirement for your application and an opportunity for you to present clearly to the assessor how you have met the CREST criteria and reflected on your project.

If you are working in a group, make sure that you fill in and submit a separate Student Profile Form for each individual.

All scientists and engineers are creative. They use scientific and technical knowledge, make decisions, solve problems, evaluate and communicate their work. Throughout your CREST project you'll need to use and show these skills too.

Managed by: British Science Association

Supported by: urenco The Energy to Succeed; UK Research and Innovation

Criteria	Where do you show this in your report or project record?	Your notes to the assessor (optional)
1 – Planning your project		
1.1 You have set a clear aim for the project and have broken it down into smaller objectives	*Section 1, Page 2; Section 5, Page 10*	*My aim is broken down to five objectives on page 2. The last paragraph of Section 5 on page 10 checks every objective against the results.*
1.2 You have explained a wider purpose for the project	*Section 1, Page 2;*	*The first two paragraphs on page 2 explain the problem as well as a personal incident that motivated me to make this project.*
1.3 You have identified a range of approaches to the project	*Section 3, Pages 4-5;*	*Three approaches with their trade-offs are considered and evaluated in a table format.*
1.4 You have described your plan for the project and why you chose that approach	*Section 3, Pages 5-8;*	*Pages 5 to 8 explain the chosen approach and the reasons for selecting it*
1.5 You have planned and organized your time well	*Section 1, page 3;*	*A table with me planned timeline along with the actual timeline is made which can be verified from the GitHub commit history. Two four-week slips were absorbed by the break I planned for my April and May exams and my SAT but and the write-up of the report went two weeks over.*
2 – Throughout your project		
2.1 You have made good use of the materials and people available	*Acknowledgements, Page 13; Appendix B, Page 15;*	*The acknowledgements list the dataset authors, papers and open-source tools I built on. Appendix B has the exact library versions and the dataset hash.*
2.2 You have researched the background to the project and acknowledged your sources appropriately	*Section 2, Page 4; References, Pages 13-14; Acknowledgements, Page 13;*	*My background reading ran from the 2009 study that created the benchmark to TypeNet in 2021. Every source is cited in the text by number in 'References'.*

3 – Finalising your project		
3.1 You have made logical conclusions and explained the implications for the wider world	*Section 6, Page 11;*	*Section 6 answers the aim directly with the measured numbers. It also covers what the result means for account security and where it should not be trusted.*
3.2 You have explained how your actions and decisions affected the project's outcome	*Section 4, Page 9;* *Section 5, Page 10;*	*Fixing the training/test leak changed every number in the report. Wiring the blended verifier into the evaluation is where the 10.2% comes from, and Appendix C shows why I kept my original settings after tuning did worse on the real test.*
3.3 You have explained what you have learnt and reflected on what you could improve	*Section 8, Page 12;*	*I learnt why biometrics are judged on error-rate curves rather than accuracy, and what working without a mentor costs when a convenient result needs doubting. The improvements are listed on page 12.*
4- Project-wide criteria		
4.1 You have shown understanding of the science behind your project, appropriate to the Award level	*Sections 2-3, Pages 4-8;* *Appendix C,* *Pages 17-19;* *Whole Report;*	*Section 2 explains how biometric systems are scored and why a single accuracy figure would be meaningless. Appendix C works through the mechanism behind each technique, like why the covariance estimate must be shrunk when only twelve enrolment samples exist in 128 dimensions.*
4.2 You have made decisions to direct the project, taking account of ethical and safety issues	*Section 7, Pages 11-12;*	*Consent is opt-in, only the fingerprint is stored, and every failure asks for another factor. The 37 times error spread between people is reported as a fairness problem.*
4.3 You have shown creative thinking	*Section 3, Page 5;*	*The central project idea is running classical statistics inside a learned fingerprint, something I came up with on my own.*

4.4 You have identified and overcome problems successfully	*Section 4, Page 9;*	*Four problems are explained it detail with their causes and fixes.*
4.5 You have explained your project clearly, in writing or conversation	*Whole report;* *Appendix A, Page 14;*	*The report is written to be followed without a technical background. Complex terminology is in the glossary and the deep detail stays in the appendices.*

Personal reflections

Now that you've finished your project, use this space to add further thoughts on what you did and evaluate each stage of the project process. The CREST Award student guide gives an example of what to include. You can continue on a separate sheet if necessary and use diagrams or pictures if you want to.
I chose this project after someone broke into my Gmail account. Luckily, I found out about the break very soon so I could secure my account before any significant damage was done. I had switched off 2FA in my account which got me thinking how easily someone with a stolen password can break into an account. While playing a typing game the idea struck me that what if I could make a system that not only looks at the password entered but also the manner in which it is typed. My project produced a 14.2% primary and 10.2% ensemble EER on 16 unseen subjects. I could not beat the 9.6% published baseline which I was hoping to exceed but I was still proud of myself and I documented the entire process in detail. Making many mistakes along the way was one of the disadvantages of not doing this project with a mentor but I learned a lot about AI through various sources including YouTube videos not only on matter directly related to my project but other concepts in the field of AI too. Going forward I plan on collecting a small consented set of real users to test the model and then based on their feedback and typing results I can improve it accordingly. I believe if I can tune this model to eventually not just cross the 9.6% EER baseline but far surpass it, the application for this project is endless and something like this truly has the potential to change the world of cybersecurity.
My mentor or supervisor
If you had a project mentor or placement supervisor (such as a teacher, relative, industry expert, etc.) how did they help with your project? (Please leave blank if you did not have one.)
Ask your mentor to confirm that this project is your work by signing below. **Signature of mentor or supervisor**
Date: 09/07/2026

Space for further notes/ drawings/ reflections (optional)

The system is live at ‘typing-sanctuary.onrender.com’. The Train page builds a typing profile in the browser and shows a similarity score when you type again, so you can watch the idea work in about two minutes.

Everything in the report can be re-run locally. The code, data pipeline and the full log of thirteen fixes are publicly available at ‘github.com/devadit1515/typing-sanctuary’.